



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

GTIIT

CALCULUS 1 - M

WINTER SEMESTER - 2024

LECTURE 21 - DECEMBER 23RD



SEQUENCES

SEQUENCES

A sequence is an ordered set

$$\{a_n : 1 \leq n, n \in \mathbb{N}\} = \{a_n\}_{n=1}^{\infty}$$

A sequence is then the **range** of a function

$$f : \mathbb{N} \rightarrow \mathbb{R} : a_n = f(n)$$

- Some sequences can be defined by giving a formula for the n -th term or by writing out the terms of the sequence.
- n can start at 1, at 0 or at any natural number.

SEQUENCES

Ex 1. Write down the first 5 terms of each sequence:

a) $\left\{ \frac{1}{n} \right\}_{n=1}^{\infty}$

b) $\left\{ \frac{(-1)^n}{4^n} \right\}_{n=1}^{\infty}$

c) $\left\{ \sqrt{2n-6} \right\}_{n=3}^{\infty}$

d) $\left\{ \cos\left(\frac{n\pi}{6}\right) \right\}_{n=0}^{\infty}$

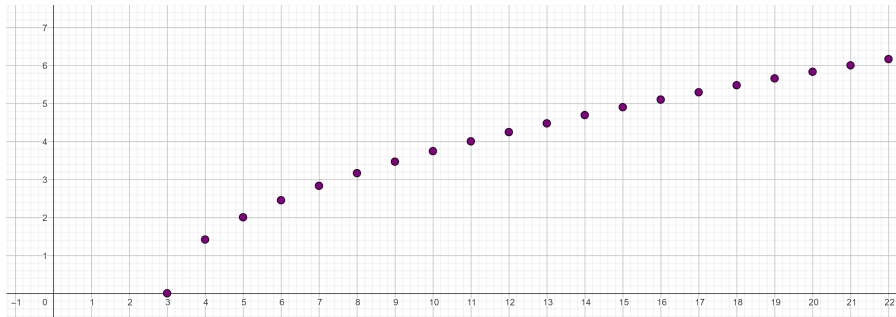
Solution a) $\left\{ \frac{1}{n} \right\}_{n=1}^{\infty} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{n}, \dots \right\}$

b) $\left\{ \frac{(-1)^n}{4^n} \right\}_{n=1}^{\infty} = \left\{ -\frac{1}{4}, \frac{1}{16}, -\frac{1}{64}, \frac{1}{256}, -\frac{1}{1024}, \dots, \frac{(-1)^n}{4^n}, \dots \right\}$

SEQUENCES

$$\text{c) } \left\{ \sqrt{2n-6} \right\}_{n=3}^{\infty} = \left\{ 0, \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \dots, \sqrt{2n-6}, \dots \right\}$$

SEQUENCE 1. c)



$$\left\{ \sqrt{2n-6} \right\}_{n=3}^{\infty} = \left\{ 0, \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \dots, \sqrt{2n-6}, \dots \right\}$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = \infty$$

SEQUENCES

1. d) $\left\{ a_n = \cos\left(\frac{n\pi}{6}\right) \right\}$

$$\bullet a_1 = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad a_2 = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}, \quad a_3 = \cos\left(\frac{\pi}{2}\right) = 0$$

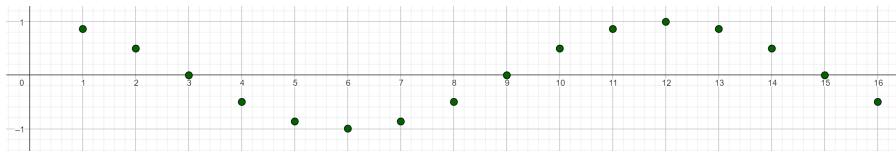
$$a_4 = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}, \quad a_5 = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$$

$$\implies \left\{ \cos\left(\frac{n\pi}{6}\right) \right\}_{n=0}^{\infty} = \left\{ \frac{\sqrt{3}}{2}, \frac{1}{2}, 0, -\frac{1}{2}, \frac{\sqrt{3}}{2}, -1, -\frac{\sqrt{3}}{2}, \dots, \cos\left(\frac{n\pi}{6}\right), \dots \right\}$$

$$\bullet a_{13} = \cos\left(\frac{13\pi}{6}\right) = \cos\left(2\pi + \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \implies a_{12k+n} = a_n \quad \forall k, n \in \mathbb{N}$$

$\implies a_n$ is periodic (of period 12) and so, it doesn't have any limit

SEQUENCE 1. D)



Graph of $\left\{ \cos\left(\frac{n\pi}{6}\right) \right\}_{n=0}^{\infty}$ in $\mathbb{R}^2$

$$\left\{ \cos\left(\frac{n\pi}{6}\right) \right\}_{n=0}^{\infty} = \left\{ \frac{\sqrt{3}}{2}, \frac{1}{2}, 0, -\frac{1}{2}, \frac{\sqrt{3}}{2}, -1, -\frac{\sqrt{3}}{2}, \dots, \cos\left(\frac{n\pi}{6}\right), \dots \right\}$$

SEQUENCES

Ex 2. Find the formula for the general term a_n of the sequence given by $\{-1, \frac{4}{9}, -\frac{5}{27}, \frac{2}{27}, -\frac{7}{243}, \dots\}$ assuming that the pattern continues in the same way

Solution. Write $a_n = \frac{b_n}{c_n}$: we see that the signs of the terms are alternately positive and negative.

• The denominator is a power of 3, propose $c_n = (-1)^n 3^n$

• $a_2 = \frac{(2+2)}{3^2}$, $a_3 = -\frac{(3+2)}{3^3} \implies$ propose the quotient and check:

$$a_n = \frac{(-1)^n (n+2)}{3^n}$$

$$a_1 = -\frac{1+2}{3^1} = -1 \checkmark \quad a_4 = \frac{4+2}{3^4} = \frac{2}{27} \checkmark \quad a_5 = -\frac{5+2}{3^5} \checkmark$$

THE FIBONACCI SEQUENCE: WAS FOUND IN THE 13-TH CENTURY SOLVING A PROBLEM CONCERNING THE BREEDING OF RABBITS.

The Fibonacci Sequence is defined recursively by the conditions

$$f_1 = 1 = f_2, \quad f_n = f_{n-1} + f_{n-2}$$

that is, each term is the sum of the two **preceding terms**:

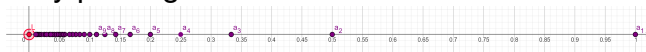
$$\{1, 1, 2, 3, 5, 8, 13, 21, 34 \dots\}$$

A SEQUENCE AS ISOLATED POINTS OF A CURVE

A sequence such as the one in Example 1. a):

$\left\{\frac{1}{n}\right\}_{n=1}^{\infty} = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{n}, \dots\right\}$ can be pictured either

- By plotting its terms on a number line



- By plotting its graph as points of a function of variable x :

$$f(x) = \frac{1}{x} \implies f(n) = \frac{1}{n}$$

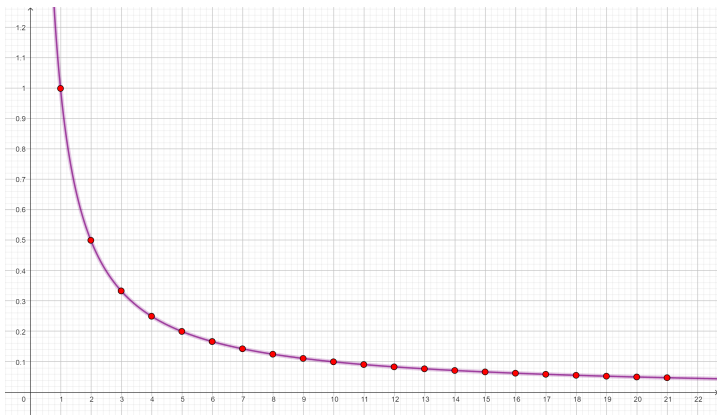
- Since a sequence is a function whose domain is the set of positive integers, its graph consists of isolated points with coordinates

$$\left\{(1, a_1), (2, a_2), (3, a_3), \dots, (n, a_n), \dots\right\}$$

A SEQUENCE AS ISOLATED POINTS OF A CURVE

- $f(x) = \frac{1}{x} \implies f(n) = \frac{1}{n}$

- $\lim_{n \rightarrow \infty} \frac{1}{n} = \lim_{n \rightarrow \infty} \frac{1}{x} = 0$



A SEQUENCE AS ISOLATED POINTS OF A CURVE

The sequence of Example 1. b):

$$\left\{ \frac{(-1)^n}{4^n} \right\}_{n=1}^{\infty}$$

can not be seen as points of a curve of the variable x

- Since any exponential function is only defined for a positive base

$$f(x) = b^x : b > 0$$

$\implies (-1)^n$ only makes sense for $n \geq 0, n \in \mathbb{Z}$, so there is no $f(x)$ such that $f(n) = a_n$ for $x = n$.

A SEQUENCE AS ISOLATED POINTS OF A CURVE

Example 1. c):

$$\left\{ a_n = \sqrt{2n-6} \right\}_{n=3}^{\infty} = \left\{ 0, \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \dots, \sqrt{2n-6}, \dots \right\}$$

can be plotted as points of a function of variable x :

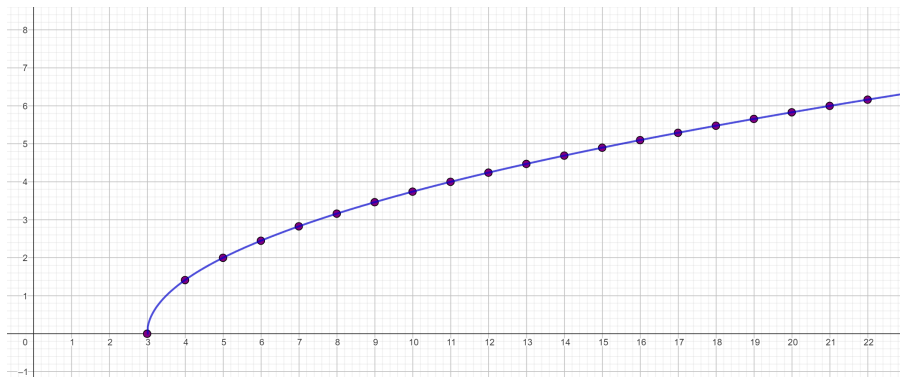
$$f(x) = \sqrt{2x-6} \implies f(n) = \sqrt{2n-6} : n \geq 3$$

The graph of $\{a_n\}$ consists of the isolated points with coordinates

$$\left\{ (1, a_1), (2, a_2), \dots \right\} = \left\{ (1, f(1)), (2, f(2)), (3, f(3)), \dots, (n, f(n)), \dots \right\}$$

$$\bullet \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt{2n-6} = \infty$$

A SEQUENCE AS ISOLATED POINTS OF A CURVE



$$\left\{ \sqrt{2n-6} \right\}_{n=3}^{\infty} = \left\{ 0, \sqrt{2}, 2, \sqrt{6}, 2\sqrt{2}, \dots, \sqrt{2n-6}, \dots \right\}$$

$$f(x) = \sqrt{2x-6} : x \geq 3 \implies f(n) = \sqrt{2n-6}$$

Ex 3. Find the limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

$$\left\{ \left(1 + \frac{1}{n} \right)^n \right\}_{n=1}^{\infty} = \left\{ a_1 = 2, a_2 = 2.25, a_3 = 2.37, a_4 = 2.44, a_5 = 2.49, a_6 = 2.52, a_7 = 2.54, a_8 = 2.56, a_9 = 2.57, a_{10} = 2.58, a_{11} = 2.59, a_{12} = 2.6, a_{13} = 2.61, a_{14} = 2.62, a_{15} = 2.63, a_{16} = 2.64, a_{17} = 2.65, a_{18} = 2.66, a_{19} = 2.67, a_{20} = 2.68, a_{21} = 2.69, a_{22} = 2.7, a_{23} = 2.71, a_{24} = 2.72, a_{25} = 2.73, a_{26} = 2.74, a_{27} = 2.75, a_{28} = 2.76, a_{29} = 2.77, a_{30} = 2.78, a_{31} = 2.79, a_{32} = 2.8, a_{33} = 2.81, a_{34} = 2.82, a_{35} = 2.83, a_{36} = 2.84, a_{37} = 2.85, a_{38} = 2.86, a_{39} = 2.87, a_{40} = 2.88, a_{41} = 2.89, a_{42} = 2.9, a_{43} = 2.91, a_{44} = 2.92, a_{45} = 2.93, a_{46} = 2.94, a_{47} = 2.95, a_{48} = 2.96, a_{49} = 2.97, a_{50} = 2.98, a_{51} = 2.99, a_{52} = 3.0, a_{53} = 3.01, a_{54} = 3.02, a_{55} = 3.03, a_{56} = 3.04, a_{57} = 3.05, a_{58} = 3.06, a_{59} = 3.07, a_{60} = 3.08, a_{61} = 3.09, a_{62} = 3.1, a_{63} = 3.11, a_{64} = 3.12, a_{65} = 3.13, a_{66} = 3.14, a_{67} = 3.15, a_{68} = 3.16, a_{69} = 3.17, a_{70} = 3.18, a_{71} = 3.19, a_{72} = 3.2, a_{73} = 3.21, a_{74} = 3.22, a_{75} = 3.23, a_{76} = 3.24, a_{77} = 3.25, a_{78} = 3.26, a_{79} = 3.27, a_{80} = 3.28, a_{81} = 3.29, a_{82} = 3.3, a_{83} = 3.31, a_{84} = 3.32, a_{85} = 3.33, a_{86} = 3.34, a_{87} = 3.35, a_{88} = 3.36, a_{89} = 3.37, a_{90} = 3.38, a_{91} = 3.39, a_{92} = 3.4, a_{93} = 3.41, a_{94} = 3.42, a_{95} = 3.43, a_{96} = 3.44, a_{97} = 3.45, a_{98} = 3.46, a_{99} = 3.47, a_{100} = 3.48, a_{101} = 3.49, a_{102} = 3.5, a_{103} = 3.51, a_{104} = 3.52, a_{105} = 3.53, a_{106} = 3.54, a_{107} = 3.55, a_{108} = 3.56, a_{109} = 3.57, a_{110} = 3.58, a_{111} = 3.59, a_{112} = 3.6, a_{113} = 3.61, a_{114} = 3.62, a_{115} = 3.63, a_{116} = 3.64, a_{117} = 3.65, a_{118} = 3.66, a_{119} = 3.67, a_{120} = 3.68, a_{121} = 3.69, a_{122} = 3.7, a_{123} = 3.71, a_{124} = 3.72, a_{125} = 3.73, a_{126} = 3.74, a_{127} = 3.75, a_{128} = 3.76, a_{129} = 3.77, 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= 5.66, a_{319} = 5.67, a_{320} = 5.68, a_{321} = 5.69, a_{322} = 5.7, a_{323} = 5.71, a_{324$$

LIMIT OF A SEQUENCE

Find the limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

$$\left\{ \left(1 + \frac{1}{n}\right)^n \right\}_{n=1}^{\infty}$$

$$= \left\{ a_1 = 2, a_2 = 2.25, a_3 = 2.37, a_4 = 2.44, a_5 = 2.49, a_6 = 2.52, a_7 = \dots \right\}$$

can be plotted as points of $f(x) = \left(1 + \frac{1}{x}\right)^x : x \geq 1$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

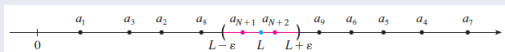
1. DEFINITION: LIMIT OF A SEQUENCE

- A sequence $\{a_n\}_{n=1}^{\infty}$ has the limit $L \in \mathbb{R}$ and we write

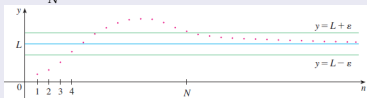
$$\lim_{n \rightarrow \infty} a_n = L$$

if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\text{if } n > N \text{ then } |a_n - L| < \epsilon$$



No matter how small an interval $(L - \epsilon, L + \epsilon)$ is chosen, there exists an N such that all terms of the sequence from a_N onward must lie in that interval.



$$\lim_{n \rightarrow \infty} a_n = L$$

2. THEOREM: LIMIT OF A SEQUENCE

If $L \in \mathbb{R}$,

$$f(n) = a_n \ \forall n \text{ and } \lim_{x \rightarrow \infty} f(x) = L \implies \lim_{n \rightarrow \infty} a_n = L$$

LIMIT OF A SEQUENCE

Ex 4. Find $\lim_{n \rightarrow \infty} \frac{\ln(n^2)}{(4n^2 - 1)}$

Solution. • Use the function $f(x) = \frac{\ln(x^2)}{(4x^2 - 1)} : x \geq 1$ such that $f(n) = \frac{\ln(n^2)}{(4n^2 - 1)} = a_n \forall n \geq 1$

• By L'Hospital's Rule for the function $f(x)$:

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{[\ln(x^2)]^{\rightarrow \infty}}{(4x^2 - 1)^{\rightarrow \infty}} = \lim_{x \rightarrow \infty} \frac{2 \ln(x)}{(4x^2 - 1)} = \lim_{x \rightarrow \infty} \frac{\left(\frac{2}{x}\right)}{8x} \\ &= \lim_{x \rightarrow \infty} \frac{1}{4x^2} = 0\end{aligned}$$

• $\lim_{x \rightarrow \infty} f(x) = L = 0 \implies \lim_{n \rightarrow \infty} a_n = 0$ ✓



3. THEOREM. LIMITS OF SEQUENCES

If $c \in \mathbb{R}$, $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent sequences, that tend to L_1 and $L_2 \in \mathbb{R}$, respectively, then the following operations are convergent sequences and:

$$1) \lim_{n \rightarrow \infty} (a_n + b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) + \left(\lim_{n \rightarrow \infty} b_n \right) = L_1 + L_2$$

$$2) \lim_{n \rightarrow \infty} (a_n - b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) - \left(\lim_{n \rightarrow \infty} b_n \right) = L_1 - L_2$$

$$3) \lim_{n \rightarrow \infty} (c \cdot a_n) = c \cdot \lim_{n \rightarrow \infty} a_n = c \cdot L_1$$

$$4) \lim_{n \rightarrow \infty} (a_n \cdot b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) \cdot \left(\lim_{n \rightarrow \infty} b_n \right) = L_1 \cdot L_2$$

$$5) \lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} = \frac{L_1}{L_2} \text{ if } \lim_{n \rightarrow \infty} b_n \neq 0$$

$$6) \lim_{n \rightarrow \infty} (a_n)^p = \left[\lim_{n \rightarrow \infty} a_n \right]^p = L^p : p > 0$$

LIMIT OF A SEQUENCE

Ex 5. a) Given $\epsilon = 10^{-1}$, $\epsilon = 10^{-3}$, find N such that

$$n > N \implies \left| \frac{1}{n} - L \right| < \epsilon$$

Solution. • Given $\epsilon = 10^{-1}$, compute

$$\left| \frac{1}{n} - L \right| = \frac{1}{n} < \frac{1}{10} \iff 10 < n.$$

So, for $\epsilon = 10^{-1}$, let $N = 10$, then

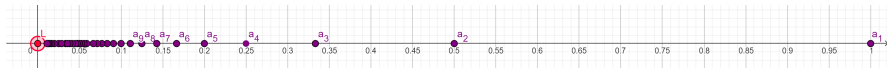
$$n > 10 \implies \left| \frac{1}{n} - L \right| = \frac{1}{n} < \frac{1}{10} < \epsilon$$

• Given $\epsilon = 10^{-3}$, compute $\left| \frac{1}{n} - L \right| = \frac{1}{n} < \frac{1}{10^3} \iff 10^3 < n$

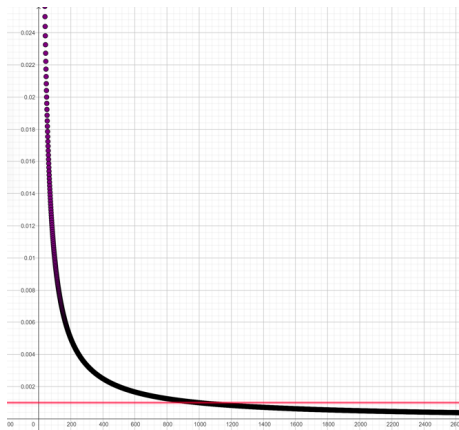
So, for $\epsilon = 10^{-3}$, let $N = 10^3$, then

$$n > 10^3 \implies \left| \frac{1}{n} - L \right| = \frac{1}{n} < \frac{1}{10^3} < \epsilon \checkmark$$

LIMIT OF A SEQUENCE



LIMIT OF A SEQUENCE



$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \text{and} \quad n > 10^3 \implies \left| \frac{1}{n} - L \right| = \frac{1}{n} < \frac{1}{10^3} < \epsilon \quad \checkmark$$

LIMIT OF A SEQUENCE

Ex 5. b) Prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Proof. Given $\epsilon > 0$, there are two possibilities:

- Case $\epsilon \geq 1$, then choose $N = 1$, since, for any $n \geq 2$:

$$n \geq 2 \implies \frac{1}{n} < \dots < \frac{1}{4} < \frac{1}{3} < \frac{1}{2} < 1 \leq \epsilon$$

So, for every $\epsilon \geq 1$, there exists $N = 1$ such that

$$n > N \implies \left| \frac{1}{n} - L \right| = \frac{1}{n} < 1 \leq \epsilon \quad \checkmark$$

- Case $0 < \epsilon < 1$: there exists an integer larger than $\frac{1}{\epsilon} > 1$, since this is true for any real number, take the smallest among them: $N = \text{Min}\{n \in \mathbb{N} : \frac{1}{\epsilon} < n\}$, so

$$n > N > \frac{1}{\epsilon} \implies \left| \frac{1}{n} - L \right| = \frac{1}{n} < \epsilon$$

So, there exists N such that if $n > N \implies \left| \frac{1}{n} - L \right| < \epsilon \quad \checkmark$

4. DEF. INFINITE LIMIT: DIVERGENCE TO ∞

- A sequence $\{a_n\}_{n=1}^{\infty}$ tends to ∞ and we write

$$\lim_{n \rightarrow \infty} a_n = \infty$$

if for every $M > 0$ there exists $N \in \mathbb{N}$ such that

$$\text{if } n > N \text{ then } a_n > M$$

- A sequence $\{a_n\}_{n=1}^{\infty}$ tends to $-\infty$ and we write

$$\lim_{n \rightarrow \infty} a_n = -\infty$$

if for every $M > 0$ there exists $N \in \mathbb{N}$ such that

$$\text{if } n > N \text{ then } a_n < -M$$

5. THEOREM: LIMIT OF SEQUENCES

- If $f(n) = a_n \forall n$, $\lim_{x \rightarrow \infty} f(x) = \infty \implies \lim_{n \rightarrow \infty} a_n = \infty$

- If f is continuous at $L \in \mathbb{R}$ and

$$\lim_{n \rightarrow \infty} a_n = L \implies \lim_{n \rightarrow \infty} f(a_n) = f(L)$$

- If $\lim_{n \rightarrow \infty} |a_n| = 0 \implies \lim_{n \rightarrow \infty} a_n = 0$

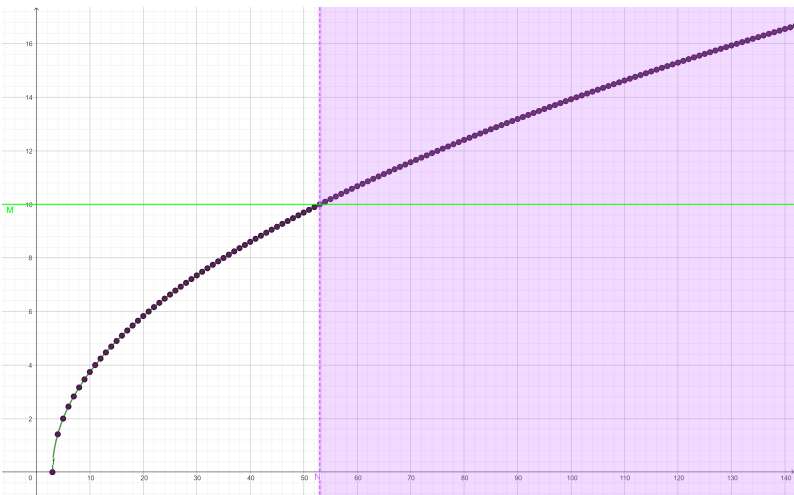
- **Squeeze Theorem.** If $a_n \leq b_n \leq c_n \forall n \geq n_0$
(not necessarily for all $n \geq 1$),

$$\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n \implies \lim_{n \rightarrow \infty} b_n = L$$

that is, $\{b_n\}_{n=1}^{\infty}$ is a convergent sequence to the same L .

INFINITE LIMIT OF A SEQUENCE

$$\lim_{n \rightarrow \infty} a_n = \infty \Leftrightarrow n > N \text{ then } \sqrt{2n - 6} > M$$



INFINITE LIMIT OF A SEQUENCE

Ex 6. a) Given $M_1 = 10$, $M_2 = 10^3$, find N such that

$$n > N \implies \sqrt{2n - 6} > M$$

Solution. • Given $M = 10$, compute

$$\begin{aligned}\sqrt{2n - 6} > 10 &\iff 2n - 6 > 10^2 \iff 2n > 106 \\ &\iff n > \frac{10^2 + 6}{2} = 53 \iff n > 53\end{aligned}$$

• So, for $M = 10$, take $N = 53$, then

$$n > 53 \implies 2n > 106 \implies 2n - 6 > 100$$

$$\implies \underbrace{\sqrt{2n - 6}}_{=a_n} > \sqrt{100} = 10 \implies \sqrt{2n - 6} > 10 \checkmark$$

$$\implies \implies \sqrt{2n - 6} > 10 \checkmark$$

INFINITE LIMIT OF A SEQUENCE

Given $M = 10^3$, find N such that $n > N \implies \sqrt{2n-6} > M$

Solution. • Given $M = 10^3$, compute

$$\sqrt{2n-6} > 10^3 \iff 2n-6 > 10^6 \iff 2n > 10^6 + 6$$

$$\iff n > \frac{10^6+6}{2} = 500,003 \iff n > 500,003$$

• So, for $M = 10^3$, take $N = 500,003$, then

$$n > 500,003 \rightarrow 2n > 10^6 + 6 \implies 2n - 6 > 10^6$$

$$\implies \underbrace{\sqrt{2n-6}}_{=a_n} > \sqrt{10^6} = 10^3 \implies \sqrt{2n-6} > 10^3 \checkmark$$

INFINITE LIMIT OF A SEQUENCE

Ex 6. b) Prove that $\lim_{n \rightarrow \infty} \sqrt{2n - 6} = \infty$

Proof. • Given $M > 0$, we wish to find N :

$$n > N \implies \sqrt{2n - 6} > M$$

• Compute $\sqrt{2n - 6} > M$

$$\iff 2n - 6 > M^2 \iff 2n > M^2 + 6 \iff n > \frac{M^2 + 6}{2}$$

Hence, given $M > 0$, take $N \geq \frac{M^2 + 6}{2}$, then

$$n > N \implies \sqrt{2n - 6} > M \quad \checkmark$$

LIMIT OF A SEQUENCE

Ex 7. Determine whether the sequence $a_n = (-1)^n$ is convergent or divergent.

Solution. • If we write out the terms of the sequence, we obtain

$$\{ -1, 1, -1, 1, -1, 1, \dots \}$$

• Since the terms **oscillate between 1 and -1 infinitely often** $\implies a_n$ **does not approach any number L .**

$\implies \lim_{n \rightarrow \infty} (-1)^n$ does not exist; that is, **the sequence $\{(-1)^n\}$ is divergent** ✓

LIMIT OF A SEQUENCE

Ex 8. Determine whether the sequence $a_n = \frac{(-1)^n}{n}$ is convergent or divergent.

Solution. • Let us compute the limit of $|a_n|$ and apply Theorem 5:

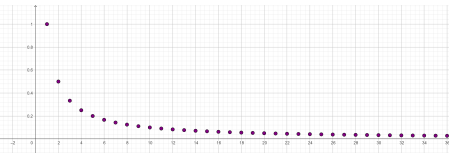
$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

• Th 5: If $\lim_{n \rightarrow \infty} |a_n| = 0 \implies \lim_{n \rightarrow \infty} a_n = 0$

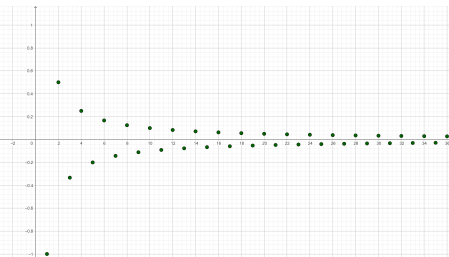
$$\implies \lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0 \quad \checkmark$$

LIMIT OF A SEQUENCE

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$$



$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{n} \right| = 0$$



$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$$

SEQUENCES

Ex. 9. Find the limit and prove it by definition:

$$a_n = \frac{2 + n + 3n^2}{7n^2}$$

Solution: $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2 + n + 3n^2}{7n^2} = \frac{3}{7}.$

Proof:

$$\left| a_n - \frac{3}{7} \right| = \left| \frac{2 + n}{7n^2} + \frac{3}{7} - \frac{3}{7} \right| = \left| \frac{2 + n}{7n^2} \right| < \frac{2n}{7n^2} < \epsilon$$

if we choose $\frac{2}{7\epsilon} < n$ and $3 \leq n$ ✓

LIMIT OF A SEQUENCE

Ex 10. Define $n! = 1 \cdot 2 \cdot 3 \cdots n : \forall n \geq 1$.

Determine whether the sequence is convergent or divergent:

$$a_n = \frac{n!}{n^n}$$

Solution. • Let us compute the first terms of $|a_n|$:

$$a_1 = 1, \quad a_1 = \frac{2!}{2^2} = \frac{1}{2} = 0.5$$

$$a_3 = \frac{3!}{3^3} = \frac{1 \cdot 2 \cdot 3}{3 \cdot 3 \cdot 3} = \frac{2}{9} \approx 0.22$$

$$a_4 = \frac{4!}{4^4} = \frac{1 \cdot 2 \cdot 3 \cdot 4}{4 \cdot 4 \cdot 4 \cdot 4} = \frac{3}{32} \approx 0.09$$

$$a_5 = \frac{5!}{5^5} = \frac{24}{625} \approx 0.04$$

$$a_6 = \frac{6!}{6^6} = \frac{5}{324} \approx 0.02$$

LIMIT OF $a_n \frac{n!}{n^n}$

$$\bullet a_n = \frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdots n}{n \cdot n \cdot n \cdots n} = \frac{1}{n} \left(\frac{2 \cdot 3 \cdots n}{n \cdot n \cdots n} \right)$$

- The expression in parentheses is **at most 1** because the numerator is **less than (or equal to) the denominator**. So

$$0 < a_n \leq \frac{1}{n}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$0 \qquad \qquad \qquad 0$$

Therefore,

$$\implies \lim_{n \rightarrow \infty} a_n = 0 \checkmark$$

MATHEMATICAL INDUCTION

Mathematical Induction is a powerful technique for proving certain types of mathematical statements: **general propositions which assert that a statement is true for all positive integers** or for all positive integers from some point on.

THEOREM. MATHEMATICAL INDUCTION

Given a the statement $P(k)$ for each $k \geq 1$, suppose that

- The base case (or initial case) $P(1)$ is true.
- The induction step (or inductive step, or step case) $P(n) \implies P(n+1)$ is true, meaning that, if assuming that the statement $P(n)$ holds for some arbitrary natural number n , it is possible to prove that the statement $P(n+1)$ holds.

Then, $P(k)$ is true for all $k \in \mathbb{N}$ ✓.

EX. PROVE BY INDUCTION $a_n = \frac{n!}{n^n} \leq \frac{1}{n} \forall n$

Proof. • The base case (or initial case) $P(1)$ is true:

$$P(1) : a_1 = \frac{1}{1} = 1 \leq \frac{1}{1} \checkmark$$

• The induction step $P(n) \implies P(n+1)$ is true:

Suppose that $a_n = \frac{n!}{n^n} \leq \frac{1}{n}$, then

$$a_{n+1} = \frac{(n+1)!}{(n+1)^{(n+1)}} = \frac{(n+1)}{(n+1)} \frac{n!}{(n+1)^n} = \frac{n!}{n^n} \frac{n^n}{(n+1)^n} = a_n \cdot \left(\frac{n}{n+1}\right)^n$$

Notice that $n < (n+1) \implies \frac{n}{n+1} < 1$, then

$$\implies a_{n+1} = a_n \cdot \left(\frac{n}{n+1}\right)^n \leq \frac{1}{n} \cdot \left(\frac{n}{n+1}\right)^n = \underbrace{\left(\frac{n}{n+1}\right)^{n-1}}_{<1} \frac{1}{(n+1)} \leq \frac{1}{(n+1)} \checkmark$$

Then, $P(k)$ is true for all $k \in \mathbb{N}$ $\checkmark$.